Graph Neural Network-Based WiFi Indoor Localization System

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***Abstract*—As mobile devices become increasingly popular and the need for indoor localization services grows, the localization of indoor mobile users is becoming more and more popular. However, the instability of received signal strength in the actual environment will have a detrimental influence on indoor localization, and the large multi-story buildings will also create new challenges. In this paper, we put forward a localization model with the graph-based location mapping network. The connection mode of the access points is used to construct a graph describing the location of reference points and users. Also, the graph neural networks are used to extract graph-level representation. This model can effectively capture the misaligned features. Moreover, the proposed approach is assessed on two public datasets, i.e., UJIIndoorLoc and UTSIndoorLoc, and the performance is evaluated against several leading-edge methods. Experimental results demonstrate that the proposed model outperforms existing solutions.**

***Index Terms*—Indoor Localization, Received Signal Strength, Graph Neural Network**

I. Introduction

With the widespread adoption of wireless mobile devices, indoor location based services (LBS), including homecare [1], healthcare [2], and monitoring [3], have attracted tremendous attention. The WiFi-based technologies [4] for LBS are most popular and can be categorized into two main groups: mathematical geometry model-based and fingerprint-based strategies. With the rapid development of neural networks, the fingerprint-based methods prevail over the geometric ones.

Fingerprint-based localization generally is composed of offline and online stages. In the former, fingerprints at reference points (RPs) with known locations are gathered to build a radio map. Note that a radio fingerprint can be a received signal strength (RSS) vector or a channel state information (CSI) vector. During the online deployment phase, positioning algorithms are used to predict the mobile users' location by matching users' RSS or CSI signals with the radio map. Compared to CSI, RSS is easier to obtain and requires fewer hardware resource, being more popular in most positioning systems. For example, Baidu and Google have deployed numerous access points (APs) in about 4,000 buildings to provide RSS fingerprint-based LBS. Within the past few years, machine learning algorithms, such as the k-nearest neighbor (KNN) [5], have been widely used in RSS fingerprint-based localization. Due to the powerful feature capturing capability, deep learning could achieve better results in a large indoor environment. In [6], Kim *et al.* used stacked autoencoders and feed-forward deep neural network (DNN) to implement the floor classification task. Song *et al.* applied a one-dimensional convolutional neural network based indoor localization (CNNLoc) for floor classification and regression [7]. In [8], Cha *et al.* proposed a location system with a hierarchical auxiliary DNN (HADNN), which consists of several consecutive feed-forward neural networks (CFNN).

Although works on fingerprint-based positioning have sprung up, some challenges still exist. The major one is the uncertainty of the RSS measurements. Since RSS is easily influenced by random factors, including unpredictable people movement, uncertain doors, and severe weather, there may exist significant distribution differences between the test samples and those in the radio map. In other words, the random environment may rapidly change, especially in large indoor scenes, and the radio maps will lose their timeliness, which causes the erroneous matching and results in poor positioning performance [9]. Correspondingly, a solution is to frequently update the radio map [10], [11]. However, due to the power outages or connection errors, the number of detected APs in offline and online stages may be different, which yields a misaligned radio map. Traditionally, some works adopted several global constants to denote the RSS values of the disappeared or emerged APs [6], [7]. These constants actually mismatch with the real environment, which deteriorates the localization performance.

In this paper, to address the aforementioned issues, we bring up a localization model using a graph-based location mapping network. The key idea about the model is that although an AP's RSS measurement at the RP may be noisy or missing, proximity patterns (describes the AP's relative position to other APs) are generally more stable. Based on the RSS values at the neighbor APs, we utilize the graph topology to fill the features at the missing AP. Then, through the graph neural network (GNN), more stable representations can be used to adapt to the dynamic environmental changes. To verify our proposed model, we apply the public UJIndoorLoc [12] and UTSIndoorLoc [7] databases.

II. Problem Formulation

In the paper, we consider a three-dimensional indoor scene, which contains several APs and RPs. Here, the element index

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vector of all APs is defined as $\boldsymbol{\alpha} = [1, 2, \ldots, N]^T \in \mathbb{R}^{N\times 1}$, and the element index vector of all RPs is given by $\boldsymbol{\gamma} = [1, 2, ..., R]^T \in \mathbb{R}^{R\times 1}$, where N and R are the numbers of APs and RPs, respectively. We assume that each RP can obtain all APs' RSS values. The processed fingerprint vector of the i-th RP at a known location can be written as

$$\boldsymbol{x}_i = [x_{i,1}, x_{i,2}, ..., x_{i,N}]^T, \tag{1}$$

where $x_{i,n}$ is the processed RSS value along the link from the n-th AP to the i-th RP. The marked position vector of this RP is expressed as $\boldsymbol{p}_i = [x_i, y_i, z_i]^T$. Note that $[\boldsymbol{p}_i]_{0:1} = [x_i, y_i]^T$ is composed of location coordinates, while $[\boldsymbol{p}_i]_2 = [z_i]$ represents the floor. Correspondingly, the radio map can be presented as $\mathcal{R} = \{\boldsymbol{x}_i, \boldsymbol{p}_i\}_{i=1}^{N_R}$. For the unknown point, i.e., the target point, its fingerprint and location vector can be given as $\boldsymbol{x}^u \notin \mathcal{R}$ and $\boldsymbol{p}^u$, respectively.

In fact, each RP cannot obtain the RSS values of all APs. Here, we initialize the missing values to zero and aim to recover them. Then, $\boldsymbol{x}_i$ is recovered as $\boldsymbol{y}_i = [y_{i,1}, ..., y_{i,M}]^T \in \mathbb{R}^{M\times 1}$. Within the deep learning framework, fingerprint-based localization comprises offline and online stages. In the former, we use the restorative radio map to train the mapping function $\boldsymbol{\Psi}_{\boldsymbol{\theta}}$, where $\boldsymbol{\theta}$ is the relevant parameter. In the latter, we could predict $\boldsymbol{p}^u$ by the trained $\boldsymbol{\Psi}_{\boldsymbol{\theta}}$.

Determine the mapping function $\boldsymbol{\Psi}_{\boldsymbol{\theta}}$: We can denote the mapping from $\boldsymbol{y}_i$ to $\boldsymbol{p}_i$ as

$$\boldsymbol{\Psi}_{\boldsymbol{\theta}} : \{\boldsymbol{y}_i\} \to \{\boldsymbol{p}_i\}. \tag{2}$$

Here, we aim to ensure the localization performance. Therefore, the optimal learning problem can be modeled as

$$(\text{P1}): \ \boldsymbol{\theta}^\star = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{N_R} \|\boldsymbol{\Xi}_\theta(\boldsymbol{y}_i) - \boldsymbol{p}_i\|. \tag{3}$$

Predict the target location $\boldsymbol{p}^u$: After obtaining $\boldsymbol{\theta}^\star$, we can determine the optimal mapping function $\boldsymbol{\Psi}_{\boldsymbol{\theta}^\star}$. Then, we can predict the target location $\boldsymbol{p}^u$ through its recovered fingerprints $\boldsymbol{y}^u$ as

$$\boldsymbol{\Psi}_{\boldsymbol{\theta}^\star} : \{\boldsymbol{y}^u\} \to \{\boldsymbol{p}^u\}. \tag{4}$$

III. Deep Learning Based Localization

As described in Section II, the problem in (3) can be divided into three steps as

$$\{\boldsymbol{x}_i\} \to \{\boldsymbol{y}_i\} \to \{\boldsymbol{p}_i\}, \tag{5}$$

where the first step reconstructs the missing features of fingerprints, and the second step completes the location mapping. From the perspective of the graph, we can treat each AP as a node, and each element of the fingerprint vector as a corresponding node feature. $\{\boldsymbol{x}_i\} \to \{\boldsymbol{y}_i\}$ means to reconstruct the graph signal. Considering the similarity of features among adjacent nodes, we apply the feature propagation (FP) algorithm [13] to supplement the missing features. $\{\boldsymbol{y}_i\} \to \{\boldsymbol{p}_i\}$ can be achieved by the GNN-based location mapping network. The overall architecture of the networks is displayed in Fig. 1.

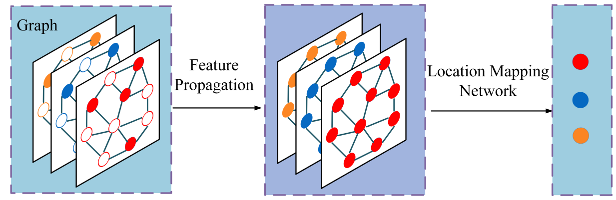

Fig. 1. The structure of the proposed localization network.

A. Graph-Based Radio Map Representation

We can use the set of graphs with labels to represent $\mathcal{R}$ as $\mathcal{R}_G = \{\mathcal{G}_i, \boldsymbol{p}_i\}_{i=1}^{N_R}$, where $\mathcal{G}_i = \{\mathcal{V}_i, \mathcal{E}_i\}$ is a weighted undirected graph. $\mathcal{V}_i$ and $\mathcal{E}_i$ represent the node and edge sets, respectively. In order to streamline the operation of $\mathcal{G}_i$, the n-th node is expressed as $v_{i,n} = x_{i,n}$. Then, we have $\mathcal{V}_i = \{v_{i,n}\}_{n=1}^N$. To elucidate the spatial association between $v_{i,n}$ and $v_{i,m}$, we define their edge as $\varepsilon_{i,n,m} = 1/d_{n,m}$, where $d_{n,m}$ is the Euclidean distance from the n-th AP to the m-th AP. Correspondingly, we have $\mathcal{E}_i = \{\varepsilon_{i,n,m}\}_{n,m=1}^{n,m=N}$. If the location of the AP is not known, we could adopt the weighted centroid localization to approximately estimate its location [14].

Moreover, adjacency matrix about $\mathcal{G}_i$ can be expressed as $\mathbf{W}_i \in \mathbb{R}^{N\times N}$, where $\mathbf{W}_i$ is a symmetric matrix, and its element $w_{i,n,m}$ reflects the connection between $v_{i,n}$ and $v_{i,m}$. If $\varepsilon_{i,n,m} > \vartheta$, $w_{i,n,m} = \varepsilon_{i,n,m}$ and $v_{i,n}$ connects with $v_{i,m}$ mutually; otherwise, $w_{i,n,m} = 0$, where ϑ is a threshold. Accordingly, the graph Laplace matrix is written as $\mathbf{L}_i = \mathbf{D}_i - \mathbf{W}_i$, where $\mathbf{D}_i = \text{diag}(\sum_{m=1}^{N} w_{i,n,m}, ..., \sum_{m=1}^{N} w_{i,n,N})$ is a degree matrix.

B. Feature Propagation

FP aims to obtain $\{\boldsymbol{y}_i\}$ from $\{\boldsymbol{x}_i\}$. Note that zero in $\boldsymbol{x}_i$ indicates a missing value. Therefore, all nodes in $\mathcal{G}_i$ can be divided into the missing node set and the normal node set. Here, we express the normal node set as $\overline{\mathcal{V}}_i \subset \mathcal{V}_i$. Theoretically, from the propagation characteristics of radio signals, $x_{i,n}$ is highly correlated with $x_{i,m}$ when the n-th AP is close to the m-th AP. This actually leads to the spatial homogeneity hypothesis. In other words, $v_{i,n}$ and $v_{i,m}$ in $\mathcal{G}_i$ are similar. Therefore, we can achieve the feature propagation between the missing nodes and the normal nodes. In [13], the Dirichlet energy $E(\mathcal{V}_i, \mathbf{L}_i)$ describes the homogeneity among nodes as

$$E(\mathcal{V}_i, \mathbf{L}_i) = \frac{1}{2}\ \boldsymbol{x}_i^T \mathbf{L}_i \boldsymbol{x}_i. \tag{6}$$

If the signal values among adjacent nodes are similar, $E(\mathcal{V}_i, \mathbf{L}_i)$ is small. Thus, the reconstruction problem can be modeled as

$$(\text{P2}): \ \arg\min_{\boldsymbol{y}_i} \frac{1}{2} \boldsymbol{y}_i^T \mathbf{L}_i \boldsymbol{y}_i, \tag{7}$$

$$\text{s.t.}\ \ y_{i,n} = v_{i,n}\ , \forall v_{i,n} \in \overline{\mathcal{V}}_i. \tag{8}$$

By solving (P2), we can obtain the missing values and restore $\mathcal{G}_i$.

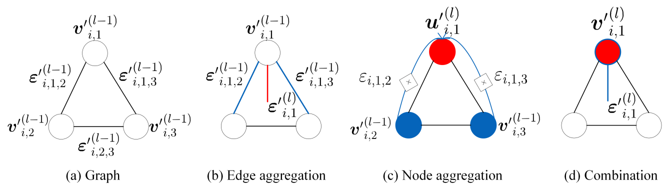

Fig. 2. Message passing. Red indicates the aggregated or combined element, and blue indicates the elements which are involved in the aggregation or combination.

C. Location Mapping Network

The location mapping network aims to obtain $\{\boldsymbol{p}_i\}$ from $\{\boldsymbol{y}_i\}$. Moreover, the input of the network is the restored graph $\mathcal{G}_i$. In this paper, a GNN-based technique is adopted to implement the location mapping network. As shown in Fig. 3, the location mapping network comprises two primary elements: 1) residual unit based message passing and 2) graph-level representation and localization.

1) Residual Unit Based Message Passing: The main unit of the localization mapping network is the message passing layer. Specifically, message passing is mainly divided into three sub-units: the edge aggregation module $f^{\varepsilon-v}$, the node aggregation module f^{v-v} and the combination module f^{v}, as shown in Fig. 2. It is worth noting that the original attribute of $v_{i,n}$ and $\varepsilon_{i,n,m}$ in $\mathcal{G}_i$ are discrete. Thus, we need to map them to the low dimensional space through the embedding operation as

$$\{\boldsymbol{v}'^{(0)}_{i,n}, \boldsymbol{\varepsilon}'^{(0)}_{i,n,m}\} = f_{\boldsymbol{\theta}_{em}}\big(\{v_{i,n}, \varepsilon_{i,n,m}\}\big), \tag{9}$$

where $f_{\boldsymbol{\theta}_{em}}$ is the embedding function about the learnable parameter $\boldsymbol{\theta}_{em}$. $\boldsymbol{v}'_{i,n} \in \mathbb{R}^{K\times 1}$ and $\boldsymbol{\varepsilon}'_{i,n,m} \in \mathbb{R}^{K\times 1}$ are separately the extended node features and edge features.

In the l-th message passing layer, $f^{\varepsilon-v}$ aggregates all the embedding values of the edges connected with $\widetilde{v}_{i,n}$ as

$$\begin{aligned}\boldsymbol{\varepsilon}'^{(l)}_{i,n} &\leftarrow f^{\varepsilon-v}\big(\{\boldsymbol{\varepsilon}'^{(l-1)}_{i,n,m} \mid v_{i,m} \in \mathcal{N}(v_{i,n})\}\big)\\ &= \sum_{v_{i,m}\in\mathcal{N}(v_{i,n})} \boldsymbol{\varepsilon}'^{(l-1)}_{i,n,m},\end{aligned} \tag{10}$$

where $\boldsymbol{\varepsilon}'^{(l)}_{i,n} \in \mathbb{R}^{K\times 1}$ is the aggregated edge.

The input of f^{v-v} consists of all the embedding values of the neighbor nodes about $v_{i,n}$ and their edges. Specially, they are aggregated as

$$\begin{aligned}\boldsymbol{u}'^{(l)}_{i,n} &\leftarrow f^{v-v}\big(\{\boldsymbol{v}'^{(l-1)}_{i,m}, \varepsilon_{i,n,m} \mid v_{i,m} \in \mathcal{N}(v_{i,n})\}\big)\\ &= \sum_{v_{i,m}\in\mathcal{N}(v_{i,n})} \varepsilon_{i,n,m}\boldsymbol{v}'^{(l-1)}_{i,m},\end{aligned} \tag{11}$$

where $\boldsymbol{u}'^{(l)}_{i,n} \in \mathbb{R}^{K\times 1}$ is the aggregated node. With $\boldsymbol{\varepsilon}'^{(l)}_{i,n}$ and $\boldsymbol{u}'^{(l)}_{i,n}$, we utilize f^{v} to combine the neighbor message and obtain the updated node embedding $\boldsymbol{v}'^{(l)}_{i,n} \in \mathbb{R}^{K\times 1}$. Furthermore, there are L message passing layers in this paper. Thus, $v_{i,n}$ can capture the information of the neighborhoods within its L hops. However, the multi-layer message passing will slow down the convergence speed and degrade $\boldsymbol{v}'^{(0)}_{i,n}$. Therefore, we utilize a residual structure and express $\boldsymbol{v}'^{(l)}_{i,n}$ as

$$\begin{aligned}\boldsymbol{v}'^{(l)}_{i,n} &\leftarrow f^{v}\big(\boldsymbol{v}'^{(l-1)}_{i,n}, \boldsymbol{u}'^{(l)}_{i,n}, \boldsymbol{\varepsilon}'^{(l)}_{i,n}\big)\\ &= \mathrm{MLP}_{\boldsymbol{\theta}_l}\big((1+\epsilon)\boldsymbol{v}'^{(l-1)}_{i,n} + \boldsymbol{u}'^{(l)}_{i,n} + \boldsymbol{\varepsilon}'^{(l)}_{i,n}\big) + \boldsymbol{v}'^{(l-1)}_{i,n},\end{aligned} \tag{12}$$

where $\mathrm{MLP}_{\boldsymbol{\theta}_l}$ is the multi-layer perceptron about the parameter $\boldsymbol{\theta}_l$, and the second item $\boldsymbol{v}'^{(l-1)}_{i,n}$ indicates the residual connection. ϵ is the learnable parameter adjusting $\boldsymbol{v}'^{(l-1)}_{i,n}$, which is helpful to better learn the difference between $\boldsymbol{v}'^{(l-1)}_{i,n}$ and $\boldsymbol{v}'^{(l-1)}_{i,m}$.

2) Graph-Level Representation and Localization: For the realization of the graph-level localization task, we need to synthesize the information of $\boldsymbol{v}'^{(L)}_{i,1}$, ..., $\boldsymbol{v}'^{(L)}_{i,M}$ and get a global graph representation $\boldsymbol{h}_{\mathcal{G}_i} \in \mathbb{R}^{K'\times 1}$. For $\mathcal{G}_i$, the contributions of each node is different. For example, some APs closer to RP are more important. Therefore, we propose an attention based graph-level representation as

$$\boldsymbol{h}_{\mathcal{G}_i} = \sum_{n=1}^{M} \frac{\exp\big(f_{\boldsymbol{\theta}_{\text{gate}}}(\boldsymbol{v}'^{(L)}_{i,n})\big)}{\sum\limits_{n'=1}^{M}\exp\big(f_{\boldsymbol{\theta}_{\text{gate}}}(\boldsymbol{v}'^{(L)}_{i,n'})\big)}\boldsymbol{v}'^{(L)}_{i,n}, \tag{13}$$

where $f_{\boldsymbol{\theta}_{\text{gate}}}$ is the multi-layer fully connected network for attention, related to the parameter $\boldsymbol{\theta}_{\text{gate}}$.

Then, $\boldsymbol{h}_{\mathcal{G}_i}$ is sent to the downstream network to get the final result. The downstream network is categorized into two divisions, i.e., the regression and classification networks, which are respectively expressed as

$$[\widehat{\boldsymbol{p}}_i]_{0:1} = f_{\boldsymbol{\theta}_{re}}(\boldsymbol{h}_{\mathcal{G}_i}) \qquad [\widehat{\boldsymbol{p}}_i]_2 = f_{\boldsymbol{\theta}_{cl}}(\boldsymbol{h}_{\mathcal{G}_i}), \tag{14}$$

where $\widehat{\boldsymbol{p}}_i$ is the estimate of $\boldsymbol{p}_i$, $f_{\boldsymbol{\theta}_{cl}}$ represents the the classification network about the parameter θ_{cl}, and $f_{\boldsymbol{\theta}_{re}}$ is the the regression network about the parameter θ_{re}.

During the off-line training stage, we minimize the mean absolute error between $\widehat{\boldsymbol{p}}_i$ and $\boldsymbol{p}_i$ to update the network parameters for the regression task. For the classification task, we use the categorical cross entropy loss function.

IV. Simulation Results

In this portion, we use the public dataset UJIIndoorLoc [12] and the UTSIndoorLoc dataset [7] to compare the performance of our model with existing models.

A. Dataset Description and Preprocessing

1) Dataset Description: The UJIIndoorLoc dataset is collected in three buildings during different periods, which consist of trainingData and validationData. Here, the maximum number of floor is 5. Moreover, the UTSIndoorLoc dataset consists of UTS_training and UTS_test, which are collected in the building with 16 floors. For the UJIIndoorLoc dataset, the number of trainingData and validationData are 19937 and 1111, respectively. We utilize trainingData to train and validate the model and define the ratio of the training set to the validation one as 7:3. Then, the validationData is

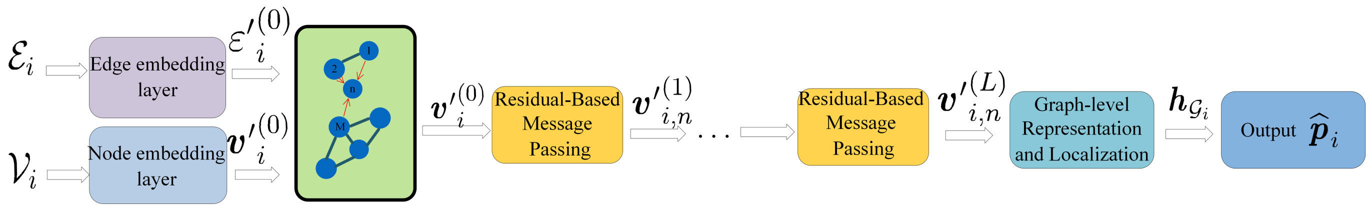

Fig. 3. The structure of the location mapping network.

used for testing. Three assignments are executed: building classification, floor classification, and latitude and longitude regression. For the UTSIndoorLoc dataset, the number of UTS_training and UTS_test are 6374 and 2733, respectively. We use UTS_training to train and validate the model, and the training set is also seven times larger than the validation set. UTS_test is used for testing. Two tasks are performed: floor classification and location coordinate regression.

2) Dataset Preprocessing: The RSS values in this dataset range from -104 dBm to 0 dBm, and the absent ones are manually marked as +100 dBm. The authors in [15] show that different manifestations of the RSS fingerprints significantly influence the accuracy of classification and the error of regression. With results in [15], we choose to standardize these RSS values and convert them into the $(0, 1)$ range as

$$x_{i,n} = \begin{cases} 0, & \text{if RSS is equal to +100 dBm} \\ \frac{RSS_{i,n} - \mu_i}{\sigma_i}, & \text{otherwise} \end{cases} \tag{15}$$

where $RSS_{i,n}$ represents the the raw RSS values from the n-th AP to the i-th RP, μ_i is the average value of the i-th RP, and σ_i is the standard deviation of the i-th RP.

B. Performance Evaluation

The parameters of the relevant layers of the location mapping network are described in TABLE I. Our architecture, implemented in Pytorch 2.9, is trained on the GPU NVIDIA GeForce RTX 3080 10 GB. The learning rate of the location mapping network η_L is set to 0.01 and the batch size N_B is 8. Moreover, the total number of epochs is 150.

We analyse the performance of the location mapping network and contrast it with the leading-edge models, including CFNN, HADNN, CNNLoc, KNN, and DNN, on the UJIIndoorLoc dataset. The comparison results of the accuracy of building, the accuracy of floor, and the mean location error are shown in TABLE III. For the building classification, most models achieve 100% success rate. For the floor classification about the UJIIndoorLoc dataset, our model outperforms most models with an accuracy of 94.15%, second only to CNNLoc. However, for the UTSIndoorLoc dataset, the accuracy of the floor classification for our model is 97.91% and is best result. In the estimation of coordinates, the mean location error of our model is significantly smaller than others, and CNNLoc has the worst result for the UJIIndoorLoc dataset. This shows that the proposed model exhibits is superior in regression tasks and classification tasks. For a qualitative evaluation,

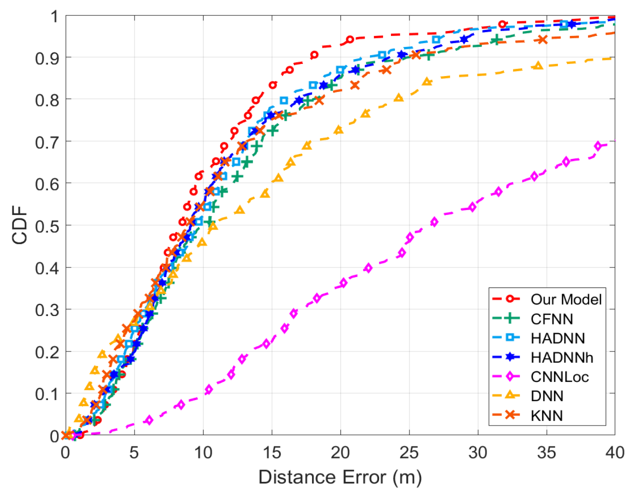

Fig. 4. The CDF of different models on the UJIIndoorLoc dataset.

the distance error curves of cumulative distribution function (CDF) are illustrated in Fig. 4 and Fig. 5. In Fig. 4, it can be evident that the curve of the location mapping network on the UJIIndoorLoc dataset is always higher than others above probability 0.4. Fig. 5 shows that the curves of CDF for our model on the UTSIndoorLoc dataset also is superior over the existing methods. It can be apparent that the curve of the location mapping network is always higher than others. This advantage over its competitors is attributed to its better representation of neighbor characteristics between APs.

V. Conclusion

In this paper, we have examined a multi-building, multi-floor, multi-point positioning system based on WiFi fingerprints, and proposed a deep learning-based positioning model, i.e., the GNN-based location mapping network, to achieve higher performance positioning. We have evaluated the proposed model on two open source datasets (i.e., UJIIndoorLoc and UTSIndoorLoc), and the experimental results have demonstrated its superiority over the existing methods.

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TABLE I
THE PARAMETERS OF THE LOCATION MAPPING NETWORK.

Layer Name	Output Size	BatchNorm1d	Activation	Dropout (0.25)
Embedding Layer	$(N_B \times M) \times 256$	No	None	No
Messing Passing Layer1	$(N_B \times M) \times 256$	Yes	ReLu	Yes
Messing Passing Layer2	$(N_B \times M) \times 256$	Yes	ReLu	Yes
Messing Passing Layer3	$(N_B \times M) \times 128$	No	LeakReLu	Yes
Graph-level Representation Layer	$(N_B \times 1) \times 128$	No	None	No
Linear Layer Based on Localization	$N_B \times 8$ *or* $N_B \times 2$	No	None	No

TABLE II
PERFORMANCE OF THE GNN-BASED LOCATION MAPPING NETWORK ON THE UTSINDOORLOC DATASET.

Layer Name	Our Model	CFNN	HADNN	HADNNh	CNNLoc	DNN	KNN
Building Accuracy	**100%**	**100%**	**100%**	99.82%	99.27%	99.10%	99.82%
Floor Accuracy	94.15%	92.34%	93.15%	93.88%	**96.03%**	79.12%	92.16%
Mean Location Error (m)	**9.61**	11.74	11.59	11.27	11.78 (34.85)	17.53	12.66

TABLE III
PERFORMANCE OF THE GNN-BASED LOCATION MAPPING NETWORK ON THE UTSINDOORLOC DATASET.

Layer Name	Our Model	CFNN	HADNN	HADNNh	CNNLoc	DNN	KNN
Building Accuracy	**100%**	**100%**	**100%**	99.82%	99.27%	99.10%	99.82%
Floor Accuracy	94.15%	92.34%	93.15%	93.88%	**96.03%**	79.12%	92.16%
Mean Location Error (m)	**9.61**	11.74	11.59	11.27	11.78 (34.85)	17.53	12.66

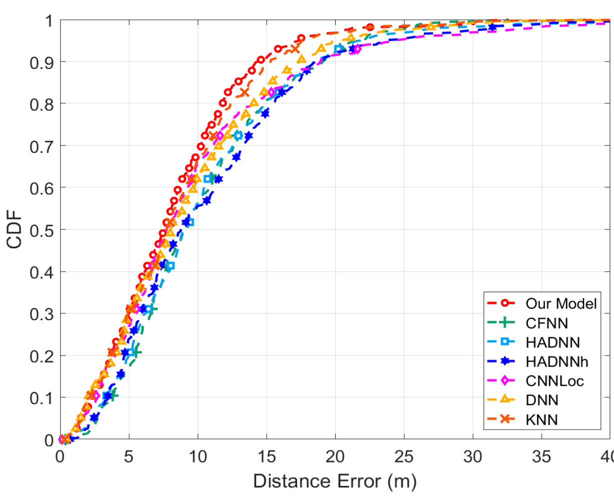

Fig. 5. The CDF of different models on the UTSIndoorLoc dataset.

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